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Non-Preemptive Shortest Job First CPU Scheduling Algorithm

Completed

Write a C program to implement the Non-Preemptive Shortest Job First (SJF) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), and Burst Times (BT). At any given time, the process with the minimum burst time among the arrived processes must be executed first. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and find the overall average TAT and WT.

Core Logic

Selection: Out of all processes that have arrived by the current time, the one with the minimum Burst Time is chosen.

Formulae:

- $\text{Turnaround Time (TAT)} = \text{Completion Time (CT)} - \text{Arrival Time (AT)}$
- $\text{Waiting Time (WT)} = \text{Turnaround Time (TAT)} - \text{Burst Time (BT)}$

Input Format

1. The first line of input contains an integer representing the number of processes.

< **Q1** >

Save

Select Language : C (GCC 9.2.0)

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5
6     printf("Enter number of processes: \n");
7     scanf("%d", &n);
8
9     int at[n], bt[n], ct[n], tat[n], wt[n];
10    int completed[n];
11
12    for(int i = 0; i < n; i++) {
13        completed[i] = 0;
14    }
15
16    for(int i = 0; i < n; i++) {
17        printf("Enter Arrival Time and Burst Time for P%d: \n", i+1);
18        scanf("%d %d", &at[i], &bt[i]);
19    }
```

Test Cases **2**[Run Sample Cases](#)[Run All Cases](#)

Custom Test Case

> Custom Test

Sample Test Case

> Test Case - 1

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```
20
21 int time = 0, count = 0;
22 float total_tat = 0, total_wt = 0;
23
24 while(count < n) {
25     int min_bt = 1000000;
26     int index = -1;
27
28     for(int i = 0; i < n; i++) {
29         if(at[i] <= time && completed[i] == 0) {
30             if(bt[i] < min_bt) {
31                 min_bt = bt[i];
32                 index = i;
33             }
34         }
35     }
36
37     if(index == -1) {
38         time++; // If no process has arrived, increment time
```

Test Cases 2

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Select Language : C (GCC 9.2.0)

```
36
37     if(index == -1) {
38         time++; // If no process has arrived, increment time
39     } else {
40         time += bt[index];
41         ct[index] = time;
42         tat[index] = ct[index] - at[index];
43         wt[index] = tat[index] - bt[index];
44
45         total_tat += tat[index];
46         total_wt += wt[index];
47
48         completed[index] = 1;
49         count++;
50     }
51 }
52
53 printf("\nAverage Turnaround Time = %.2f\n", total_tat / n);
54 printf("Average Waiting Time = %.2f\n", total_wt / n);
```

Test Cases 2

Run Sample Cases

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```
51     }
52
53     printf("\nAverage Turnaround Time = %.2f\n", total_tat / n);
54     printf("Average Waiting Time = %.2f\n", total_wt / n);
55
56     return 0;
57 }
```

[Test Cases](#) 2[Run Sample Cases](#)[Run All Cases](#)

Custom Test Case

▼ Custom Test

Success ✓

Input :

```
4
0 6
0 8
0 7
0 3
```

Expected Output :

```
Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:
Enter Arrival Time and Burst Time for P4:
```

```
Average Turnaround Time = 13.00
Average Waiting Time = 7.00
```

Output :